MILITARY SPECIFICATION

CANISTER, SMOKE, WP, FOR 152 MILLIMETER CARTRIDGE, XM410E1

1. SCOPE

1.1 This specification covers the filling of a canister with white phosphorus (WP) for a 152 millimeter (mm) cartridge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-F-320 - Fiberboard; Sheet, Stock and Cut Shapes.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-A-2550 - Ammunition and Special Weapons, General Specifications For.

FSC 1320

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

PICATINNY ARSENAL

P-119987 - WP Loading Assembly.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain that all materials and components listed on Drawing P-119987 have met their respective specifications prior to their use.

3.2 Filling. The canister, clean and free from foreign matter, shall be filled with molten WP. When water is in contact with the WP during the filling operation, the water used shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. The depth of water remaining on the surface of the WP in the filled and closed container shall not exceed 1/8 inch. The filling shall be solidified with the canister in the filling-end-up position (see 6.2 and 6.3).

3.3 Assembly. The canister shall be assembled in accordance with Drawing P-119987.

3.4 Filling height. When examined as specified in 4.6.1, the surface of the WP in the filled and closed canister shall be 3 to 3-7/32 inches from the top of the canister. The canister shall be in the filling-end-up position and the WP at a temperature of 130° ± 10°F.

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3.5 Cleaning after closing. The exterior surface of the filled and closed canister and the interior of the threaded area shall be thoroughly cleaned to remove any grease, oil, WP or foreign matter (see 6.4).

3.6 Leakage. The canister shall not leak when maintained at a temperature of 155° to 160°F for a minimum period of 8 hours and tested as specified in 4.6.2. After the canister has been accepted for leakage, the WP shall be allowed to solidify in the filling-end-up position.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will insure equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 Inspection substations. If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

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4.3 Lotting. A lot shall consist of the canisters filled by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.5 Inspection provisions.

4.5.1 Examination and nondestructive tests. Sample canisters shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.5.2 Critical defects. Each canister in the lot shall be inspected for critical defects listed in the classification of defects.

4.5.4 Classification of defects.

4.5.4.1 WP loading assembly (Dwg P-119987).

Categories	Defects	Acceptance standards
Critical:		
1	Cleaning after closing	4.6.2
2	WP leakage	4.6.3
Major:	AQL 1.0 percent defective	
101	Marking incorrect	
102	Excess water in canister	4.6.1
103	Canister not thoroughly cleaned after filling and closing (see 3.5)	
Major A:	AQL 0.65 percent defective	
150	Filling height excessive	4.6.1
Minor:	AQL 2.5 percent defective	
201	Filling height insufficient	4.6.1
202	Marking missing	

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4.6 Tests.

4.6.1 Filling height. The filling height and depth of water remaining on the surface of the WP filled canister shall be determined radiographically with the canister in the filling-end-up position. Edgewood Arsenal Inspection Standard R13-33 shall be used to determine conformance.

4.6.2 Cleaning after closing. The method for cleaning shall be a procedure approved by the Government inspector for conformance to 3.5.

4.6.3 Leakage. The filled and closed canister shall be placed in an oven at the specified temperature for the required period of time. The canister shall be oriented in the filling-end-down or in the horizontal position.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging, level A. After filling, the threaded surfaces of the canister shall be coated with type P-7 preservative oil, conforming to MIL-P-116. Excess oil shall be wiped off. A plastic or metal threaded shipping cap shall be placed on the threaded end of each filled canister.

5.2 Packing. Level shall be as specified (see 6.1).

5.2.1 Level A. *Twenty-four filled canisters preserved and packaged* as specified in 5.1 shall be packed with the threaded end up, in one layer of four rows, six per row in an overseas type or class container, conforming to PPP-B-576, PPP-B-585, PPP-B-601 or PPP-B-621. The box shall be fitted with full depth interlocking fiberboard partitions, forming 24 snug-fitting individual cells. Fiberboard filler pads shall be placed between the canisters and the sides, ends, top and bottom of the box. Additional fiberboard filler pads shall be used as required to make a tight pack. The fiberboard partitions and filler pads shall be fabricated from type CF, class-weather resistant, variety SW, grade optional of PPP-F-320. The box shall be closed and strapped in accordance with the requirements of the applicable container specification and the appendix thereto. Galvanized strapping shall be used. The data card shall be in accordance with MIL-A-2550.

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5.2.2 Level B. Twenty-four filled canisters, preserved and packaged as specified in 5.1, shall be packed in the same manner as specified in 5.2.1 except that a domestic type or class container and coated-finish strapping shall be used.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129. Color code identification marking shall be as shown on figure 1.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Supplier need not furnish radiographic equipment.
(c) The Government will furnish the inspection standard.
(d) Level of packing required.

6.2 Caution. Due to the hazardous nature of WP, filling installations are advised to take all possible precautions to protect personnel from injury while filling canisters.

6.3 Filling aid. The canister contains approximately 3.5 pounds of WP when filled. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 Cleaning. The filled canister may be cleaned by passing through a hot water and detergent spray washer or by flushing with hot water while vigorously scrubbing with steel wool. Additional cleaning of the threaded area may be effected with a steam lance and a power driven wire brush.

6.5 Safety classification. This item is chemical group C, explosive hazard class 2 (Army Materiel Command Regulation No. 385-235).

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1320-A033

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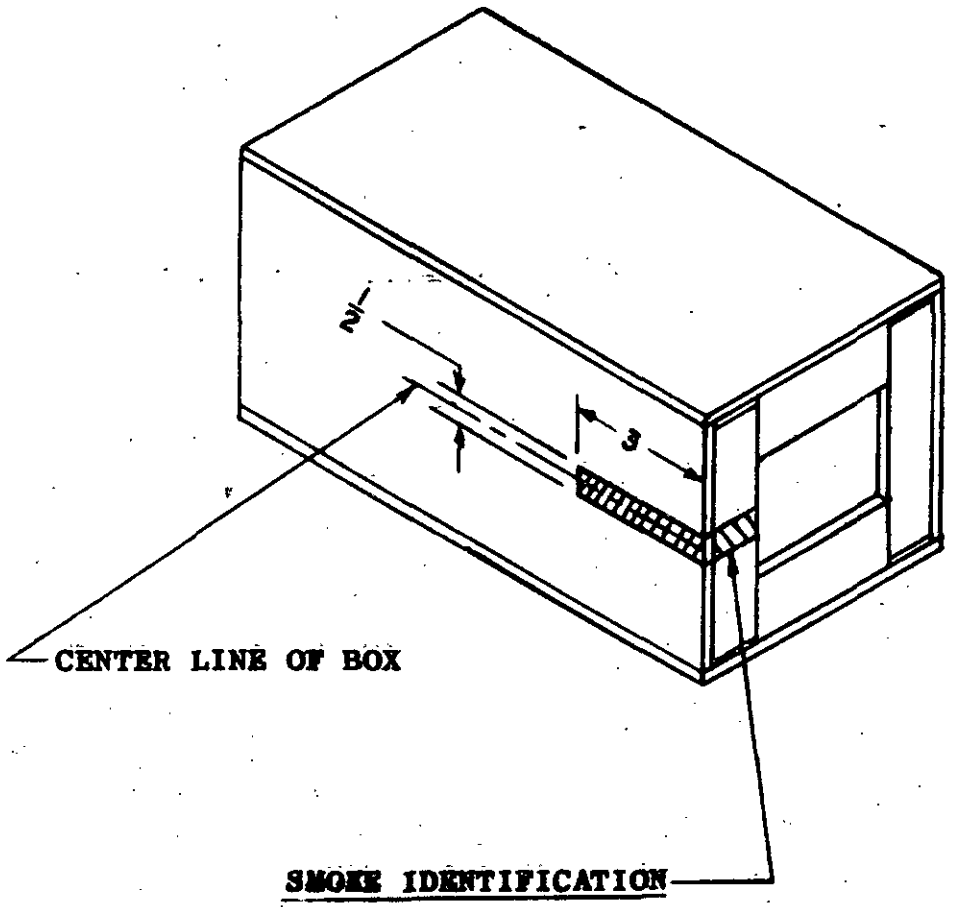


Figure 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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